DEPI

Web-to-Warehouse Product Intelligence Pipeline

Digital Egypt Pioneers Initiative (DEPI) · Graduation Capstone Project 2026

⚡ Business Need & Solution

Retail Competitive Intelligence Automation

⚠️ Problem Statement

Retailers struggle with slow, manual competitor product tracking resulting in inconsistent records, missing currency standards, duplicate listings, and zero price snapshot history.

✨ Proposed Solution

Automated, compliance-first ETL pipeline that harvests structured product feeds, resolves duplicates, models historical price changes into a Kimball star warehouse, and orchestrates with Airflow.

14,918
Canonical Products (Real)

86,788
Historical Price Snapshots

100%
Robots.txt & Terms Compliant

12 / 12
Automated DQ Rules

🔄 9-Stage ETL Pipeline & Architecture

Robots-Gated Extraction to Kimball Star-Schema Analytics

1	EXTRACT	Scrape 34+ permitted sources with BeautifulSoup/Scrapy, resp
2	CLEANSE	29 cleaning rules: strip promotional noise, title-case, normalize
3	RESOLVE	Fuzzy entity matching & token fingerprinting to merge duplicat
4	MODEL	Load into PostgreSQL & MySQL Kimball warehouse: dim_produ
5	DETECT	SQL diff engine flags price hikes, drops, category drift, new arri
6	QUALITY	12-rule validation framework checks nulls, negative prices, fres

SYSTEM ARCHITECTURE FLOW

1. Ingestion Layer
34+ APIs, Scrapy, BeautifulSoup, Rate Limiter

2. Storage
PostgreSQL 16 & MySQL

Python 3.12 · FastAPI · PostgreSQL 16 · MySQL 8.4 · Apache Airflow 2.10 · Docker & Compose · React 19 + TypeScript · BeautifulSoup4 · SQLAlche